Machine Learning and Pattern Recognition
Laboratory Report

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May 3, 2026

Abstract

This report details the implementation and analysis of dimensionality reduction techniques (PCA and LDA) and Generative Models (Multivariate Gaussian Classifiers) applied to the provided datasets.

Contents

1 Gaussian Density Estimation **2**

1 Gaussian Density Estimation

What do you observe?

When superimposing the Maximum Likelihood (ML) estimated 1D Gaussian probability density functions over the normalized histograms, we observe that the fundamental assumption of normality (that the data follows a symmetric, bell-shaped curve) is highly inconsistent across the dataset. Because the Gaussian model forces a perfectly symmetric shape centered on the mean, it behaves predictably: it wraps nicely around symmetric data but completely fails to capture the true density of data that is heavily skewed or asymmetric.

Are there features for which the Gaussian densities provide a good fit?

Yes, the Gaussian densities provide a reasonably good fit for features that naturally exhibit a roughly symmetric distribution.

- **Fake Class:** Features 1, 2, 3, and 4 provide a solid fit. The estimated red Gaussian curves successfully capture the central density peaks and the tapering tails of the empirical blue histograms.
- **Genuine Class:** Features 3 and 4 show an acceptable, albeit slightly imperfect, alignment with the Gaussian fit.

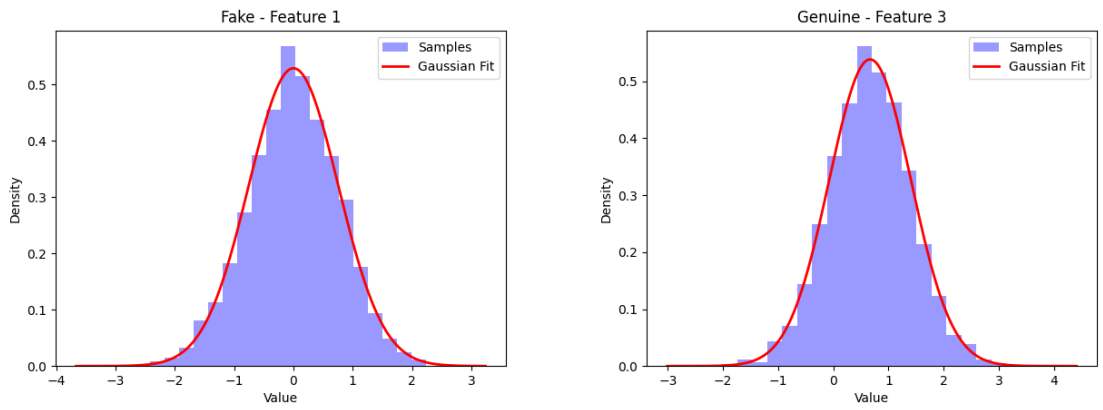


Figure 1: Examples of good Gaussian fits: Fake Feature 1 (left) and Genuine Feature 3 (right). Both exhibit a relatively symmetric, bell-shaped empirical distribution.

Are there features for which the Gaussian model seems significantly less accurate?

Absolutely. The Gaussian model is highly inaccurate for features that exhibit strong asymmetry (skewness), as the mathematical model simply cannot bend to fit a skewed shape.

- **Fake Class:** Features 5 and 6 are heavily skewed to the left. The symmetric Gaussian curve underestimates the sharp right-sided peaks and overestimates the density on the steep drop-off.
- **Genuine Class:** The inaccuracy is even more pronounced here. Features 1, 2, 5, and 6 exhibit severe negative skewness (long left tails with very sharp, sudden drop-offs on the right). For instance, the model predicts a high probability of finding samples on the far right side where essentially zero empirical samples exist.

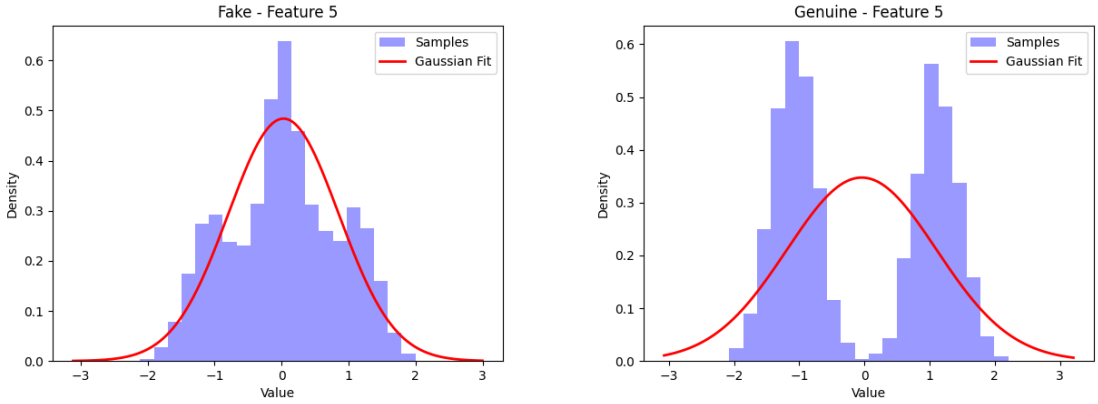


Figure 2: Examples of inaccurate Gaussian fits: Fake Feature 5 (left) and Genuine Feature 5 (right). The rigid symmetry of the Gaussian model completely misses the sharp peaks and asymmetric tails.

This visual analysis indicates that while a Gaussian classifier is mathematically elegant and computationally fast, its rigid structural assumptions will likely cause classification errors when evaluating test samples based on these heavily skewed features.